

pandas Essentials

`df.head()` · `df.info()` · `df.describe()` · `df.value_counts()` · `df[df.col > x]`. Master them and 80% of pandas feels obvious.

Before we touch a row, what is the one thing you run on EVERY new DataFrame?

L3

WHAT'S INSIDE

- 01** Series and DataFrame — the two...
- 02** Three ways to construct
- 03** The five-command inspection discipline
- 04** Access cheat sheet — `.loc` vs `.iloc` vs...
- 05** `SettingWithCopyWarning` — a real bug...

WHY THIS LESSON MATTERS

Every data analyst in India uses these five commands daily.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Construct a DataFrame from dict, list of dicts, and CSV.

2

Access rows and columns with `.loc`, `.iloc`, and masks.

3

Inspect a DataFrame with `head`, `info`, `describe`, `value_counts`.

4

Add, rename, drop columns without silent bugs.

5

Recognise `SettingWithCopyWarning` as a real bug vs noise.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Series and DataFrame — the...

A DataFrame is a dict-of-Series that share an index

02

Three ways to construct

Know when each is appropriate

03

The five-command inspection...

Run before any analysis

04

Access cheat sheet — .loc...

.loc uses labels. .iloc uses positions.
Never mix.

01

PART 1 OF 4

Series and DataFrame — the...

A DataFrame is a dict-of-Series that share an index

CONCEPT 1 OF 4

Series and DataFrame — the two primitives

Series = 1-D labelled array. Dict + array fused.

DataFrame = 2-D labelled table.

Index = row labels. Default: RangeIndex.

Tidy data: row = observation, column = variable.

MENTAL MODEL

Tidy grid

If rows aren't observations, fix the shape first.

DEFINITION

Series and DataFrame — the two...

A DataFrame is a dict-of-Series that share an index

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Three ways to construct

`pd.DataFrame({"a":[1,2],"b":[3,4]})` — dict of lists (teaching, fixtures).

`pd.DataFrame([{"a":1}, {"a":3}])` — list of dicts (API responses).

`pd.read_csv("data.csv")` — production start.

All three produce the same shape; pick the most ergonomic for your source.

RULE

Match the source

Fighting your input format costs hours.

KEY POINTS

- `pd.DataFrame({"a":[1,2],"b":[3,4]})` — dict of lists (teaching, fixtures)
- `pd.DataFrame([{"a":1}, {"a":3}])` — list of dicts (API responses)
- `pd.read_csv("data.csv")` — production start
- All three produce the same shape; pick the most ergonomic for your source

CONCEPT 3 OF 4

The five-command inspection discipline

`df.head(5)` — first five rows.

`df.info()` — dtypes, non-null counts, memory.

`df.describe()` — numeric stats.

`df["col"].value_counts()` — categorical frequencies.

`df.isna().sum()` — missing counts per column.

RITUAL

5 commands

Skipping these is the #1 source of downstream bugs.

DEFINITION

The five-command inspection discipline

Run before any analysis

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Access cheat sheet — .loc vs .iloc vs mask

.loc uses labels. .iloc uses positions. Never mix.

KEY POINTS

- loc uses labels. .iloc uses positions. Never mix
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Build, inspect, mutate — on a dataset of your choice

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which of the five inspection commands do you skip most often?

Q2

Have you ever shipped a `SettingWithCopyWarning` to production?

Q3

What one pandas habit from today will you carry forward?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

`df["a"]` vs `df[["a"]]` — differ by:

A

Nothing

B

Shape: Series vs 1-col DataFrame

C

Speed

D

Memory

df["a"] vs df[["a"]] — differ by:



CORRECT ANSWER

Shape: Series vs 1-col DataFrame

WHY

`df["a"]` returns a Series (1-D); `df[["a"]]` returns a 1-column DataFrame (2-D)

Access row by LABEL 3:

A

`df[3]`

B

`df.iloc[3]`

C

`df.loc[3]`

D

`df.row(3)`

Access row by LABEL 3:



CORRECT ANSWER

`df.loc[3]`

WHY

loc is label-based; .iloc is position-based

SettingWithCopyWarning arises from:

A

Unicode

B

Chained indexing

C

Large datasets

D

Missing values

SettingWithCopyWarning arises from:

CORRECT ANSWER



Chained indexing

WHY

Chained indexing (e.g. `df[mask]["col"] = x`) makes pandas unsure whether it's a view or a copy

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Series = 1-D labelled array. Dict + array fused.

DataFrame = 2-D labelled table.

Index = row labels. Default: RangeIndex.

Tidy data: row = observation, column = variable.

MENTAL MODEL

Tidy grid

If rows aren't observations, fix the shape first.

THE TAKEAWAY

"You already think in spreadsheets. Pandas is those same moves, made reproducible."

WHAT TO NOTICE

1

Series and DataFrame — the...

2

Three ways to construct

3

The five-command inspection...

4

Access cheat sheet — .loc...

Mini assignment — surprise me

Load any CSV you own (Kaggle counts). Run the 5-command inspection block. Report ONE surprising finding.

DELIVERABLES

- Pick a CSV with $\geq 10,000$ rows.
- Run the five inspection commands.
- Write one sentence describing the most surprising thing you found.
- Post screenshot + sentence on the cohort forum.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Screenshot + one-sentence finding.

ONE THING TO REMEMBER

“

"You already think in spreadsheets. Pandas is those same moves, made reproducible."

— AI Learning Hub

END OF LESSON 2

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 4 — Data Cleaning. Missing values, duplicates, dtypes.